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"""
Basic unit tests for the model
training/validation module.
Demonstrates the testing strategy
referenced in the Week 3 plan
(programming practices: code review,
version control, testing).

Run with: pytest
test_model_training.py
"""

import numpy as np
import pandas as pd
from sklearn.datasets import
make_classification
from model_training_validation import
(
    get_candidate_models,
    train_and_compare,
    validate_model,
    select_best_model,
)

def _sample_data():
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X, y =
make_classification(n_samples=200,
n_features=6, random_state=42)
    return pd.DataFrame(X),
pd.Series(y)

def test_candidate_models_returned():
    models = get_candidate_models()
    assert "LogisticRegression" in
models
        assert "RandomForest" in models

def
test_train_and_compare_produces_score
s():
    X, y = _sample_data()
    models = get_candidate_models()
    results =
train_and_compare(models, X, y, cv=3)
    for name, res in results.items():
        assert 0.0 <=
res["cv_f1_mean"] <= 1.0

def
test_select_best_model_returns_valid_
choice():

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        X, y = _sample_data()
        models = get_candidate_models()
        results =
train_and_compare(models, X, y, cv=3)
        best_name, best_model =
select_best_model(results)
        assert best_name in models

def
test_validate_model_metrics_in_range(
):
    X, y = _sample_data()
    models = get_candidate_models()
    results =
train_and_compare(models, X, y, cv=3)
    _, best_model =
select_best_model(results)
    metrics =
validate_model(best_model, X, y)
    for v in metrics.values():
        assert 0.0 <= v <= 1.0

```